A GENERAL FORM OF THE DIMENSION FORMULA FOR A LOCALLY FINITE TYPE MORPHISM OVER A LOCALLY NOETHERIAN BASE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finiteness of the irreducible components through a point and their description by minimal primes. Schemes, Lemma \ref{schemes-lemma-specialize-points} and Algebra, Lemma \ref{algebra-lemma-Noetherian-irreducible-components}, cited at proof lines 13629-13634. Makes the maximum over the components well defined and finite.
- A morphism into a reduced scheme factors through the closure of the image of a point. Schemes, Lemma \ref{schemes-lemma-map-into-reduction}, cited at proof lines 13640-13642. Lets the dimension formula be applied to the component over the closure of the image.
- The dimension formula for the local rings at a point and its image. Lemma \ref{lemma-dimension-formula}, cited at proof lines 13642-13647. Supplies the inequality on each component.

Source reference. Original lemma 0BAE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14207–14207. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01J7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2004–2004. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00FR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 581–581. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0356: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4117–4117. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02JU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3349–3349. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let S be a scheme. Let $f : X \to S$ be a morphism of schemes. Let $x \in X$, and set $s = f(x)$. Assume S is locally Noetherian and f is locally of finite type, We have*

$$\dim(\mathcal{O}_{X,x}) \leq \dim(\mathcal{O}_{S,s}) + E - \mathit{trdeg}_{\kappa(s)}\kappa(x). \tag{0.2.1}$$

where E is the maximum of $\mathit{trdeg}_{\kappa(f(\xi))}(\kappa(\xi))$ where ξ runs over the generic points of irreducible components of X containing x.

Proof. Let $X_1, \ldots, X_n$ be the irreducible components of X containing x endowed with their reduced induced scheme structure. These correspond to the minimal primes $\mathfrak{q}_i$ of $\mathcal{O}_{X,x}$ and hence there are finitely many of them (Schemes, Lemma 01J7 and Algebra, Lemma 00FR). Then $\dim(\mathcal{O}_{X,x}) = \max \dim(\mathcal{O}_{X,x}/\mathfrak{q}_i) = \max \dim(\mathcal{O}_{X_i,x})$. The ξ's occurring in the definition of E are exactly the generic points $\xi_i \in X_i$. Let $Z_i = \overline{\{f(\xi_i)\}} \subset S$ endowed with the reduced induced scheme structure. The composition $X_i \to X \to S$ factors through Z_i (Schemes, Lemma 0356). Thus we may apply the dimension formula (Lemma 02JU) to see that $\dim(\mathcal{O}_{X_i,x}) \leq \dim(\mathcal{O}_{Z_i,x}) + \mathrm{trdeg}_{\kappa(f(\xi))}(\kappa(\xi)) - \mathrm{trdeg}_{\kappa(s)}\kappa(x)$. Putting everything together we obtain the lemma. □